hexens

SMART CONTRACT AUDIT REPORT FOR NFTIME

15.12.2021

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Summary

NFTime's smart contracts have low risk severity. During the audit our specialists found one medium risk and multiple low risk weaknesses.

The weaknesses described in this report were fixed in the following commit: (f67364f8b5b0b2711b54fe8a906b8e934c5e7d81)

Severity	Number of Findings
High	0
Medium	1
Low	7
Informational	0

Total: 8

Scope

The analyzed contracts are located in the following repository:
https://github.com/NFTime-social/ido-smart-contract/blob/main/contracts/IDONFTIME.sol
(commit 39bd06d3fb75325ee3fde867c693e7a233940e75)

Description

This audit covered the following contracts of NFTime:

- IDONFTIME.sol

Test Cases

Testing results are presented below.

```
NFTime

    ✓ should initialize with correct values
    ✓ should revert if amount == zero
    ✓ should revert if amount > limit
    ✓ should revert if the amount is not sufficient to buy
even one NFTM token
    ✓ should revert if Public sale has already finished
    ✓ should revert if buyer does not have enough acclaimed
BUSD or allowance
    ✓ should buy the right amount of tokens and unlock
PUBLIC_IMMEDIATE_UNLOCK_PERCENTAGE(%) (45ms)
    ✓ should revert if public sale is still in progress
    ✓ [!] should revert if public supply has been drained
    ✓ should revert if not enough time has passed after public
sale has ended (46ms)
    ✓ should revert if user has never bought tokens
    ✓ should revert if user wants to withdraw twice during the
same month (50ms)
    ✓ should withdraw correct amount of tokens if the lock
duration has passed (65ms)


  12 passing (3s)
  1 failing
```

Weaknesses

This section contains the list of discovered weaknesses.

1. Clean Code Recommendation/Gas Optimisation

Severity: Low

Resolution: remove buyerAddress from structure

Description:

The **buyerAddress** field in the structure **PSBuyer** on line 32 in the file **/NFTime-social/ido-smart-contract/contracts/IDONFTIME.sol** is excessive, since the structure is accessed through address mapping (line 89).

```
struct PSBuyer {
    address buyerAddress;
    uint256 lastWithdraw;
    uint256 initialTotalBalance;
    uint256 balance;
    uint256 busdLimit;
  }
```

2. Excessive Function Call

Severity: Low

Resolution: remove the excessive check

Description:

The **require()** function call on line 68 in the file **/NFTime-social/ido-smart-contract/contracts/IDONFTIME.sol** seems excessive, since the modifier **contractNotStarted** on line 66 has the same functionality.

```
function initialize()
    external
    onlyOwner
    contractNotStarted
  {
    require(_contractStarted == false, "IDO: The IDO contract has been already initialized");
```

3. Variable Visibility Not Set

Severity: Low

Resolution: state the variable visibility

Description:

The visibility of the **psBuyers** variable on line 89 in the file **/NFTime-social/ido-smart-contract/contracts/IDONFTIME.sol** is not set. Since the visibility is set in case of other variables, it is recommended to set it for the above mentioned variable as well.

```
mapping (address => PSBuyer) psBuyers;
```

4. Excessive Function Call

Severity: Low

Resolution: remove the excessive call

Description:

The **require()** function call on line 107 in the file **/NFTime-social/ido-smart-contract//contracts/IDONFTIME.sol** is excessive, since the code on the lines 104 and 106 already guarantees the correct execution.

```
require(busdAmount > 0, "IDO: Provided BUSD amount must be higher than 0");
    ....
require(busdAmount <= psBuyers[buyer].busdLimit, "IDO: The Provided BUSD amount exceeds allowed spend limit");
require(psBuyers[buyer].busdLimit != 0, "IDO: This address has already purchased tokens for 500 BUSD");
```

5. Clean Code Recommendation/Gas Optimisation

Severity: Low

Resolution:

Description:

The variable's **memory** location defined on line 126 in the file **/NFTime-social/ido-smart-contract//contracts/IDONFTIME.sol** is recommended to be changed to **storage.** Further in code, **psBuyer** can be used to change the storage state (instead of **psBuyer[buyer]**).

```
PSBuyer memory psBuyer = psBuyers[buyer];
```

6. Clean Code Recommendation

Severity: Low

Resolution: reuse the constant

Description:

The constant used on line 132 in the file **/NFTime-social/ido-smart-contract//contracts/IDONFTIME.sol** should be rewritten to **PUBLIC_LOCK_DURATION_IN_MONTHS**.

```
if(monthsSinceDate >= 3)
```

7. State Update Failure

Severity: Medium

Resolution:

Description:

On line 186 in the file **/NFTime-social/ido-smart-contract//contracts/IDONFTIME.sol** the **withdrawLeftPublicTokens()** function does not update the value of **_publicSale.supply**. In this case, it should be set to zero, or the **pause()** function should be called, so there will be no opportunity to call the **buyTokens()** function that only checks if **_publicSale.supply > 0**. A situation can arise in which users will be able to buy tokens from the team share pool or tokens locked in the contract.

```
function withdrawLeftPublicTokens() external onlyOwner returns (bool) {
```

8. Excessive Function Call

Severity: Low

Resolution: remove the excessive function call

Description:

The **require()** function calls on lines 236 and 237 in the file **/NFTime-social/ido-smart-contract//contracts/IDONFTIME.sol** are excessive, since the **safeTransferFrom()** function will revert the transaction if the corresponding checks fail.

```
require(_busd.allowance(buyer, address(this)) >= busdToPay, "IDO: Not enough allowance to perform transfer. Please be sure to approve sufficient tokens amount");

require(_busd.balanceOf(buyer) >= busdToPay, "IDO: Not enough BUSD available in buyer's balance. Please be sure to provide sufficient BUSD amount");
...
 _busd.safeTransferFrom(buyer, address(this), busdToPay);
```

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